Management and Monitoring of Future Communication Networks: Exercise 02

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Exercise 2: Implementation of Random Access for LoRaWAN

- In the following, the example considerations of Exercise 1.1 and Exercise 1.2 are extended to a real scenario **simulating LoRaWAN** message transmissions. An example code to achieve collisions from overlapping intervals is in WueCampus.
- The time on air T_m to transmit LoRa messages can be calculated as follows
  - $T_m = T_s \cdot n_{message}$
  - Duration to transmit **a single symbol** $T_s = \frac{2^{SF}}{BW}$, with SF as the spreading factor and 125kHz bandwidth (BW)
  - The message is calculated with $n_{message} = P + 4.25 + n_{payload}$ with a preamble P = 8 (symbols)
  - With $n_{payload} = 8 + \max(\lceil n_{packet} \rceil \cdot (CR + 4), 0)$ with a coding rate CR = 4
  - $n_{packet} = \frac{(8PL - 4SF + 8 + 16CRC + 20H)}{4(SF - 2DE)}$ with PL as payload, SF as spreading factor, cyclic redundancy check CRC=1, an enabled header H=1, and a disabled low datarate optimization DE=0
- In the following, the payload for each message is **randomly chosen** between 1B and 51B
- **Exercise 2.1**
  - Calculate and plot the time on air distribution for all payloads in the range of [1B, 51B] and SF={7,8,9,10,11,12}
  - Simulate the transmission of 300 messages using spreading factor (SF=7). Repeat the simulation 500 times and plot the collision probability for the 500 re-runs
  - Repeat the same simulation **changing only the spreading factor** to 10 and 12, respectively and compare the collision probability for all 300 message transmissions and the 500 re-runs.
    What is the reason for the achieved results? Which conclusions can be drawn if Slotted ALOHA must be used?